

Visual Proof of Model Memorization: A Comparative Case Study of Dataset Overlap in Meta AI and Google Gemini

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Abstract—This paper presents empirical evidence of dataset overlap and model memorization across state-of-the-art text-to-image models, focusing on Meta AI and Google Gemini. By analyzing a complex, highly specific prompt, we demonstrate a structural similarity of over 90% in the outputs of competing models, including identical anatomical mapping, composition, and background assets. We contrast these findings with outputs from ChatGPT, Grok, and Dola AI to highlight the anomaly, discussing the implications for dataset deduplication, copyright, and AI safety triage.

Index Terms—Artificial Intelligence, Model Memorization, Dataset Overlap, AI Safety, Machine Learning.



abstract

1 INTRODUCTION

Text-to-image models have revolutionized generative art. However, the origin of training datasets and the phenomenon of “memorization” (overfitting to training samples) remain critical areas of research...

2 THE PROMPT AND VISUAL ANALYSIS

We executed the following identical prompt across five production models:

“A dramatic Wild West scene at sunset. A cowboy riding a horse made of glowing computer circuit boards and binary code. The horse is rearing up on its hind legs.”

3 DISCUSSION: DATASET OVERLAP VS. CREATIVE GENERATION

While models like ChatGPT, Grok, and Dola AI exhibited distinct creative paths, Meta AI and Gemini converged onto a nearly identical spatial layout...

4 CONCLUSION AND AI BUG BOUNTY BOUNDARIES

This behavior was reported to major AI safety teams. While traditional bug bounty programs classify model content and dataset overlaps as “out of scope,” we argue that...